

Pattern Recognition Assignment

Solving the N-Queens Problem with Exhaustive Search and Genetic Algorithms

Authors:

Akhil Nair

Raja Hashim Ali

Project Description

This project presents a comparative study of different algorithms used for solving the N-Queens problem. The implemented approaches include:

- Exhaustive Depth-First Search (DFS)
- Greedy Hill Climbing
- Simulated Annealing
- Genetic Algorithms

The algorithms were implemented in Python and evaluated using runtime and memory analysis for different board sizes. Workflow diagrams, performance graphs, and experimental comparisons were included as part of the study.

Overleaf Project Link

<https://www.overleaf.com/3893171673jntysjrwmhv#c9db52>

GitHub Repository Link

<https://github.com/ascknair/NQueens-Algorithms>

Included Materials

The repository and Overleaf project contain:

- Python source code
- Workflow diagrams
- Runtime comparison graphs
- Memory usage graphs
- Experimental results
- IEEE-format research paper
- Generated figures in vector PDF format

Technologies Used

- Python 3.13
- Matplotlib
- Graphviz
- LaTeX (Overleaf)
- GitHub

Submission Note

All implementations, graphs, workflow diagrams, and experimental results used in the report are included in the provided repository and Overleaf project for reproducibility and reference.